

Going Continuous: Implementing CM-TS for Continuous-Arm Adaptive Conjoint

This guide assumes the discrete (finite-pool) version is already built — a Bayesian logistic model for whether a design beats a *fixed competitor*, $\Pr(y=1) = \sigma(\beta_0 + \beta^\top \mathbf{z})$, with a Laplace posterior and Thompson Sampling (TS) over the pool — and specifies only what continuity changes. The continuous algorithm is **CM-TS** (continuous-manifold TS): global variance-inflated logistic TS over the kNN-validity set $\mathcal{M}$, human-only (no AI surrogate, no PPI). (We retain “manifold” only in the acronym and as informal intuition; formally $\mathcal{M}$ is a *set* — a finite union of ball-cap intersections — not a manifold.) Each round plays the single *global* maximizer of the Thompson-sampled utility over $\mathcal{M}$,

$$\mathbf{z}_t = \arg \max_{\mathbf{z} \in \mathcal{M}} \tilde{\beta}_t^\top \mathbf{z},$$

with no local search, radius, or resampling. Under an *exact* maximization oracle (**Algorithm 1**, the SOCP global max of App. A), Theorem 1 gives cumulative beat-the-competitor regret $\tilde{O}(d^{3/2}\sqrt{T})$. The proof (App. C) *adapts* the standard linear-TS optimism/resample coupling of [Agrawal and Goyal, 2013, Abeille and Lazaric, 2017] together with the GLM self-normalized/confidence machinery of [Abbasi-Yadkori et al., 2011, Filippi et al., 2010, Fauray et al., 2020] (the two linear-TS papers supply the coupling pattern, not a standalone logistic theorem). The practical/simulation implementation (**Algorithm 1'**, App. A) approximates that arg max with per-round error ϵ_t , entering the bound additively ($+\frac{1}{4} \sum_t \epsilon_t$).

1 What continuity changes, and the one object it forces

Moving the arm from “one of $M=228$ word-indices” to a free vector $\mathbf{z} \in \mathbb{R}^d$ raises two issues; one object — the kNN-validity set $\mathcal{M}$ as the action set — handles both.

- **Challenge A — unbounded search, only-local linear model.** Over all of $\mathbb{R}^d$ the objective $\tilde{\beta}^\top \mathbf{z}$ has *no* finite maximizer, and the linear logit fits only *near* the data. *Fix:* search over the compact kNN-validity set $\mathcal{M}$. The maximizer over $\mathcal{M}$ then exists, and the journal’s goodness-of-fit study shows the linear-in-embedding model is adequate *across* $\mathcal{M}$ at $d \in [16, 32]$; the regret theorem holds for the global arg max over $\mathcal{M}$ directly.
- **Challenge B — most of $\mathbb{R}^d$ decodes to garbage.** A random $\mathbf{z} \in \mathbb{R}^d$ renders incoherently; only a thin region of real-word-like embeddings renders. *Fix (same object):* $\mathcal{M}$ is the kNN sublevel set over the 228 anchors — $\mathbf{z}$ is valid iff close to real words (coherent interpolations).

Two cheap theory refinements (keep on): a *variance-inflated* Thompson draw and a *norm projection* of the estimator (§3). Laplace and the beat-the-competitor logistic carry over verbatim.

2 Setup and key objects

In: vocabulary $\mathcal{V} = \{w_1, \dots, w_M\} + \text{T5 embeddings}$; prompt p (“a sneaker, {w}”); encoder E ; generator G (SD3.5-Large); fixed competitor image x_{comp} (the benchmark to beat); human labeler L_{ora} ; horizon T ; hyperparameters (§4). **Out:** posterior $(\hat{\beta}_T, \Sigma_T)$, winner $\hat{\mathbf{z}}^*$, trajectory $\{(\mathbf{z}_t, y_t)\}$.

symbol	meaning (default)
$\mathbf{z} \in \mathbb{R}^d$	a design = a PCA-reduced T5 token (replaces the pool index)
a, β_0	the fixed competitor’s design; intercept $\beta_0 = -\beta^\top a$ absorbs its appeal
$\beta, (\hat{\beta}_t, \Sigma_t)$	design-feature weights; their Laplace posterior (mean, covariance). In the g_t -projection and Theorem 1, β denotes the <i>augmented</i> $(\beta_0, \beta) \in \mathbb{R}^{d+1}$ with feature $\phi(\mathbf{z}) = (1, \mathbf{z}^\top)^\top$, so $\{\ \beta\ \leq S\}$ bounds the intercept coordinate too
$\tilde{\beta}$	one variance-inflated Thompson draw $\sim \mathcal{N}(\hat{\beta}_{t-1}, v^2 \Sigma_{t-1})$
$\mathcal{M}$	kNN-validity set = kNN sublevel set (the action set; compact, not a manifold); $V(\mathbf{z}) = \mathbf{1}\{\mathbf{z} \in \mathcal{M}\}$
d, M, k	reduced dim (16); vocabulary (228); kNN neighbors (10)

Fixed competitor as anchor. Each round the human compares the design against the *same* competitor and picks; $y_t = 1$ iff the design beats it. The competitor is a fixed reference a , so its appeal folds into the intercept: $\sigma(\beta^\top(\mathbf{z} - a)) = \sigma(\beta_0 + \beta^\top \mathbf{z})$, $\beta_0 = -\beta^\top a$. The optimum $\mathbf{z}^* = \arg \max_{\mathbf{z} \in \mathcal{M}} \beta^{*\top} \mathbf{z}$ (the design most likely chosen over the competitor) is unchanged by a . The regressor is the design $\mathbf{z}_t$ itself (with intercept), *not* a difference.

PCA-T5 arm space. Encode each word to $\mathbb{R}^{4096}$; PCA the M centered embeddings; keep top $d=16$ loadings U_d and mean $\bar{\mu}$; reduced anchors $\mathbf{z}_i := U_d^\top(\tilde{\mathbf{z}}(w_i) - \bar{\mu})$. The survey manipulates $\mathbf{z} \in \mathbb{R}^d$; the anchors *define* $\mathcal{M}$.

Validity oracle / kNN-validity set. With $d_k(\mathbf{z})$ the distance to the k -th nearest anchor (exact NN search over $\{\mathbf{z}_i\}$ in the d -dim PCA space; instant at $M=228$), calibrate τ_d once as the 95th percentile of leave-one-out within-vocabulary k -NN distances, and set

$$\mathcal{M} := \{\mathbf{z} : d_k(\mathbf{z}) \leq \tau_d\} = \bigcup_{|\mathcal{I}|=k} \bigcap_{i \in \mathcal{I}} B(\mathbf{z}_i, \tau_d), \quad V(\mathbf{z}) = \mathbf{1}\{\mathbf{z} \in \mathcal{M}\}.$$

So $\mathbf{z}$ is valid iff at least k anchors lie within τ_d ($\mathcal{I} \subseteq [M]$, $|\mathcal{I}| = k$, indexes a k -subset of anchors; the norm bound S is unrelated). $\mathcal{M}$ is *compact* (a finite union of closed, bounded ball-cap intersections), which is all the regret proof requires of it (App. C, Step 0). This union-of-ball-cap form makes the acquisition a finite set of convex programs (§3).

3 The main loop (CM-TS)

Variance-inflated Thompson draw. Draw $\tilde{\beta} \sim \mathcal{N}(\hat{\beta}, v^2 \Sigma)$ with exploration scale v ($v=1$ is vanilla TS and what works in practice; the worst-case guarantee uses $v = \Theta(\sqrt{d \log T})$). Split the draw into intercept and design weights, $\tilde{\beta} = (\tilde{\beta}_0, \tilde{\mathbf{b}})$ with $\tilde{\mathbf{b}} \in \mathbb{R}^d$. Since $\phi(\mathbf{z}) = (1, \mathbf{z}^\top)^\top$, the intercept is constant in $\mathbf{z}$ and drops out of the acquisition, so the best design under the draw maximizes the *design-only* form $\tilde{\mathbf{b}}^\top \mathbf{z}$ over $\mathcal{M}$ (equivalently $\tilde{\beta}^\top \phi(\mathbf{z})$).

Acquisition = exact global arg max over $\mathcal{M}$ (SOCP). Since $\mathcal{M} = \bigcup_{|\mathcal{I}|=k} \bigcap_{i \in \mathcal{I}} B(\mathbf{z}_i, \tau_d)$ exactly, the global max is the best regional max, and each region's max is a *second-order cone program* (SOCP) [Boyd and Vandenberghe, 2004] ($\max_{\mathbf{z}} \tilde{\mathbf{b}}^\top \mathbf{z}$ s.t. $\|\mathbf{z} - \mathbf{z}_i\| \leq \tau_d$, on the design subvector $\tilde{\mathbf{b}}$). The pairwise screen $\|\mathbf{z}_i - \mathbf{z}_j\| \leq 2\tau_d$ is *necessary* for a region to be non-empty, hence *lossless*; SOCP infeasibility drops the (pairwise-close but Helly-)empty ones. So enumerating *all* pairwise-screened k -subsets and solving one SOCP each (Algorithm 1, `cvxpy`/ECOS) returns the *exact* global maximizer over $\mathcal{M}$ (details, App. A). The enumeration is finite but *worst-case combinatorially infeasible* (e.g. at $M=228, k=10$, depending on τ_d): the pairwise screen is *necessary but not sufficient* for region feasibility, so it need not prune the count to tractable size. The implementation therefore runs the cheaper Algorithm 1' (per-round gap = ϵ_t), and Theorem 1 applies to the deployed acquisition only up to the additive $\frac{1}{4} \sum_t \epsilon_t$, provided that cumulative gap is controlled (App. A).

Algorithm 1 GLOBALARGMAX($\tilde{\mathbf{b}}, \{\mathbf{z}_i\}, \tau_d, k$) $\rightarrow \mathbf{z}_t$: *exact* global maximization of $\tilde{\mathbf{b}}^\top \mathbf{z}$ over $\mathcal{M}$ (design subvector $\tilde{\mathbf{b}} \in \mathbb{R}^d$; the oracle Theorem 1 analyzes)

- 1: $\mathcal{R} \leftarrow \emptyset$
 - 2: **for every** pairwise-screened k -subset $\mathcal{I}$ ($\|\mathbf{z}_i - \mathbf{z}_j\| \leq 2\tau_d \ \forall i, j \in \mathcal{I}$; lossless screen) **do**
 - 3: solve SOCP $\max_{\mathbf{z}} \tilde{\mathbf{b}}^\top \mathbf{z}$ s.t. $\|\mathbf{z} - \mathbf{z}_i\| \leq \tau_d \ \forall i \in \mathcal{I}$; if feasible, add optimizer $\mathbf{z}_{\mathcal{I}}$ to $\mathcal{R}$ (else drop: empty region)
 - 4: **end for**
 - 5: **return** $\arg \max_{\mathbf{z} \in \mathcal{R}} \tilde{\mathbf{b}}^\top \mathbf{z}$ $\triangleright$ best feasible point = global max over $\mathcal{M}$
-

Posterior update + norm projection. Render $\mathbf{z}_t$, show it against the competitor, record y_t , append $(\mathbf{z}_t, y_t)$, and refit the logistic posterior over (β_0, β) by the same Laplace update. Let $\tilde{\beta}_t$ be the unconstrained ridge quasi-MLE. Project it onto $C = \{\|\beta\| \leq S\}$ in the mean-map (g_t -)norm, $\hat{\beta}_t = \arg \min_{\beta \in C} \|g_t(\beta) - g_t(\tilde{\beta}_t)\|_{V_t^{-1}}$ with $g_t(\beta) = \lambda\beta + \sum_{s \leq t} \sigma(\beta^\top \phi(\mathbf{z}_s))\phi(\mathbf{z}_s)$ and

augmented feature $\phi(\mathbf{z}) = (1, \mathbf{z}^\top)^\top$ (App. C). Here $\beta = (\beta_0, \beta) \in \mathbb{R}^{d+1}$ is the *augmented* vector, so $C = \{\|\beta\| \leq S\}$ bounds the intercept coordinate too. This is the g_t -norm projection, *not* Euclidean [Filippi et al., 2010] (the contraction in Lemma 1 step (ii) uses the mean-map norm). In practice clipping $\|\hat{\beta}_t\| \leq S$ is the cheap stand-in — a heuristic relaxation *not* covered by Theorem 1, which is proved for the g_t -projection: swapping it for a Euclidean clip changes the confidence width $\beta_t(\delta)$ of Lemma 1, an *estimator* error distinct from (and not absorbed by) the *acquisition* gap ϵ_t of Algorithm 1'. The two relaxations are separate; see the scope note in App. B.

Generator round-trip. $\tilde{\mathbf{z}}_t = U_d \mathbf{z}_t + \bar{\mu}$; $\text{image}_t = G(\text{concat}(E(p), \tilde{\mathbf{z}}_t))$ (suffix-inject $\tilde{\mathbf{z}}_t$ as a soft token). Show it against x_{comp} ; the labeler returns $y_t = 1$ iff the design beats it.

Algorithm 2 CM-TS main loop (calls the acquisition oracle once per round)

```

1:  $(\hat{\beta}_0, \Sigma_0, \mathcal{D}_0) \leftarrow \text{INITIALIZE}(\{\mathbf{z}_i\}, n_0, L_{\text{ora}})$  ▷ §4
2: for  $t = 1, \dots, T$  do
3:    $\hat{\beta}_t = (\hat{\beta}_{0,t}, \tilde{\mathbf{b}}_t) \sim \mathcal{N}(\hat{\beta}_{t-1}, v^2 \Sigma_{t-1})$ ,  $\tilde{\mathbf{b}}_t \in \mathbb{R}^d$  ▷ variance-inflated draw; split off design subvector  $\tilde{\mathbf{b}}_t$ 
4:    $\mathbf{z}_t \leftarrow \text{GLOBALARGMAX}(\tilde{\mathbf{b}}_t, \{\mathbf{z}_i\}, \tau_d, k)$  ▷ design subvector only (intercept constant in  $\mathbf{z}$ ); exact Alg. 1, sim Alg. 1'
5:   render imaget;  $y_t \leftarrow L_{\text{ora}}(\text{image}_t, x_{\text{comp}})$  ▷ pick: design vs. competitor
6:    $\mathcal{D}_t \leftarrow \mathcal{D}_{t-1} \cup \{(\mathbf{z}_t, y_t)\}$ ;  $(\hat{\beta}_t, \Sigma_t) \leftarrow \text{Laplace update}$ ;  $\hat{\beta}_t \leftarrow g_t\text{-projection of } \tilde{\beta}_t \text{ onto } \{\|\beta\| \leq S\}$  ▷ Thm 1 estimator; [practice: Euclidean clip  $\|\hat{\beta}_t\| \leq S$ ; see §3 — not covered by Thm 1]
7: end for
8: return  $\hat{\beta}_T, \Sigma_T$ , and incumbent best played design  $\hat{\mathbf{z}}^* = \arg \max_t \hat{\mathbf{b}}_T^\top \mathbf{z}_t$  ▷  $\hat{\mathbf{b}}_T$  the design subvector of  $\hat{\beta}_T$ ; or GLOBALARGMAX( $\hat{\mathbf{b}}_T, \dots$ ) for the posterior-optimal design over  $\mathcal{M}$ 
```

4 Hyperparameters, initialization, pitfalls

param	default / formula	param	default / formula
d	16 (PCA dims)	v	1 (exploration scale)
k	10 (kNN neighbors)	S	$\ \beta\ $ bound (norm projection)
τ_d	95th pct LOO k -NN dist (§2)	n_0	24 warm-start designs

Initialization. (1) Draw n_0 random anchor designs; (2) render each, show against the competitor, record $y_i = L_{\text{ora}}$; (3) Laplace-fit $(\beta_0, \hat{\beta}_0, \Sigma_0)$ and project $\|\hat{\beta}_0\| \leq S$.

Pitfalls.

- **Search in the reduced space.** Run NN search and the SOCPs on the d -dim PCA anchors, *not* the raw 4096-dim embeddings, or τ_d and every $V(\mathbf{z})$ are wrong.
- **Regressor is the design, not a difference.** The datum is $\mathbf{z}_t$ (with intercept); $y=1$ means the design beat the competitor. Don't subtract anything — the competitor lives in β_0 .
- **Enumerate only non-empty ball-cap regions.** The exact oracle (Algorithm 1) enumerates *all* pairwise-screened k -subsets ($\|\mathbf{z}_i - \mathbf{z}_j\| \leq 2\tau_d \forall i, j$) — a *lossless* screen — and lets SOCP infeasibility drop the (Helly-)empty ones; this is exact. The pairwise screen already prunes most of the $\binom{M}{k}$ subsets, so blind enumeration is unnecessary. Any coarser *cluster* prefilter (keeping only dense anchor clusters) is an *approximation* that can drop a valid pairwise-screened region, so it belongs with the approximate Algorithm 1', not the exact oracle.
- **Fixed competitor and image seed.** Use the same competitor image (and image-generation (diffusion) seed) every round: a changing competitor shifts β_0 , a varying diffusion seed makes a design render differently — both corrupt the fit. Set the sampling RNG once.

5 Two concrete tasks

Task 1 — Build/calibrate the validity oracle (offline). Encode the 228 words $\rightarrow \mathbb{R}^{4096}$, PCA $\rightarrow d=16$ (store $U_d, \bar{\mu}, \{\mathbf{z}_i\}$); set τ_d = the 95th percentile of LOO k -NN distances. *Check coverage:* interpolated anchor pairs give $V=1$ with coherent renders (e.g. *leather* $\rightarrow$ *neon*), random interior

points mostly give $V=0$, and the LOO histogram has no isolated anchors. *Check the acquisition:* for a random draw with design subvector $\tilde{\mathbf{b}}$, run $\text{GLOBALARGMAX}(\tilde{\mathbf{b}}, \dots)$ and confirm the returned $\mathbf{z}$ has $V=1$ and beats a coarse anchor grid on $\tilde{\mathbf{b}}^\top \mathbf{z}$.

Task 2 — Run in simulation with a DreamSim oracle (known ideal). Fix an ideal image x_{ref} (known only to the simulator) and competitor x_{comp} ; replace the human with $y_t = 1$ iff $\text{DreamSim}(\text{image}_t, x_{\text{ref}}) < \text{DreamSim}(x_{\text{comp}}, x_{\text{ref}})$. After a (β_0, β) -recovery sanity check, INITIALIZE ($n_0=24$) and run T rounds; *report* the convergence curve $\text{DreamSim}(\text{image}(\mathbf{z}_t), x_{\text{ref}}) \rightarrow 0$, the running win-probability $\sigma(\hat{\beta}_t^\top \mathbf{z}_t)$, and the final winner $U_d \hat{\mathbf{z}}^* + \bar{\mu}$ beside x_{ref} .

Appendix (does not count toward the implementation-guide page limit)

A Exact-argmax computation and the simulation approximation

Exact oracle (Algorithm 1, SOCP). Write the kNN-validity set as a union of ball-cap regions, $\mathcal{M} = \bigcup_{|\mathcal{I}|=k} \bigcap_{i \in \mathcal{I}} B(\mathbf{z}_i, \tau_d)$ (exact: a point is valid iff $\geq k$ anchors lie within τ_d). Write the draw as $\tilde{\boldsymbol{\beta}}_t = (\tilde{\beta}_{0,t}, \tilde{\mathbf{b}}_t)$ with design subvector $\tilde{\mathbf{b}}_t \in \mathbb{R}^d$; since $\boldsymbol{\phi}(\mathbf{z}) = (1, \mathbf{z}^\top)^\top$, the intercept is an additive constant in $\mathbf{z}$ and

$$\max_{\mathbf{z} \in \mathcal{M}} \tilde{\boldsymbol{\beta}}_t^\top \boldsymbol{\phi}(\mathbf{z}) = \tilde{\beta}_{0,t} + \max_{\mathbf{z} \in \mathcal{M}} \tilde{\mathbf{b}}_t^\top \mathbf{z},$$

so the acquisition $\mathbf{z}_t = \arg \max_{\mathbf{z} \in \mathcal{M}} \tilde{\boldsymbol{\beta}}_t^\top \boldsymbol{\phi}(\mathbf{z}) = \arg \max_{\mathbf{z} \in \mathcal{M}} \tilde{\mathbf{b}}_t^\top \mathbf{z}$ depends only on $\tilde{\mathbf{b}}_t$. It is then a finite family of second-order cone programs [Boyd and Vandenberghe, 2004]: for each region $\mathcal{I}$, solve $\max_{\mathbf{z}} \tilde{\mathbf{b}}_t^\top \mathbf{z}$ s.t. $\|\mathbf{z} - \mathbf{z}_i\| \leq \tau_d \ \forall i \in \mathcal{I}$ (a cone solver, e.g. ECOS via `cvxpy`); drop infeasible regions; return the best feasible optimizer. Over a *single* ball $B(\mathbf{z}_i, \tau_d)$ the maximizer is the closed form $\mathbf{z}_i + \tau_d \tilde{\mathbf{b}}_t / \|\tilde{\mathbf{b}}_t\|$ (boundary point in the gradient direction). The pairwise screen $\|\mathbf{z}_i - \mathbf{z}_j\| \leq 2\tau_d \ \forall i, j \in \mathcal{I}$ is *necessary* for $\bigcap_{i \in \mathcal{I}} B(\mathbf{z}_i, \tau_d) \neq \emptyset$, hence *lossless*: enumerating **all** pairwise-screened k -subsets (the empty ones, pairwise-close but Helly-empty in $d \geq 2$, are returned infeasible by the SOCP and dropped) and solving one SOCP each returns the *exact* global maximizer over $\mathcal{M}$. Per-round cost $O(\#\text{regions} \cdot \text{SOCP}(d, k))$ (the design SOCP is over $\mathbf{z} \in \mathbb{R}^d$); the enumeration is finite but worst-case combinatorially large (dense clouds $\rightarrow$ many screened, mostly-infeasible subsets), which is why the synthetic study runs the cheaper Algorithm 1' (gap = ϵ_t). This is the oracle Theorem 1 analyzes.

Algorithm 1' — continuous approximate oracle (what the simulation runs). Exact region enumeration is overkill for the synthetic study, so the simulator uses the cheaper continuous optimizer of Algorithm 1'. It does *not* optimize over a fixed candidate pool (that would collapse CM-TS to a finite-arm bandit): with $\tilde{\mathbf{b}}_t$ the design subvector of the draw, the per-ball boundary maximizers $\mathbf{z}_i + \tau_d \tilde{\mathbf{b}}_t / \|\tilde{\mathbf{b}}_t\|$ *move with $\tilde{\mathbf{b}}_t$ every round*, and the initial candidate is refined by projected-gradient ascent onto $\partial\mathcal{M}$, so the played design is a continuous function of $\tilde{\mathbf{b}}_t$ on the continuum $\mathcal{M}$. Per-round cost $O(Md)$; its per-round gap to the exact maximizer of Algorithm 1 is the optimization error $\epsilon_t \geq 0$ that Theorem 1 absorbs additively ($+\frac{1}{4} \sum_t \epsilon_t$). The pseudocode below was written to match `cmts_sim/cmts_sim.py` (functions `argmax_over_M` and `project_to_M`) line-for-line; the code is the ground truth and the two agree.

Algorithm 1' APPROXARGMAX($\tilde{\mathbf{b}}, \{\mathbf{z}_i\}, k, \tau_d$) $\rightarrow \mathbf{z}_t$: *approximate* continuous acquisition on the design subvector $\tilde{\mathbf{b}} \in \mathbb{R}^d$ (the simulation's oracle; gap to Algorithm 1 = ϵ_t)

```

1:  $u \leftarrow \tilde{\mathbf{b}} / \|\tilde{\mathbf{b}}\|$ ;  $\mathbf{z}_i^{\text{bd}} \leftarrow \mathbf{z}_i + \tau_d u$  for all  $i$  ▷ per-ball boundary maximizers; move with  $\tilde{\mathbf{b}}$ 
2:  $\mathcal{C} \leftarrow \{\mathbf{z}_i^{\text{bd}} : d_k(\mathbf{z}_i^{\text{bd}}) \leq \tau_d\} \cup \{\mathbf{z}_i\}_{i=1}^M$  ▷ keep valid boundary points; anchors as fallback
3:  $\mathbf{z} \leftarrow \arg \max_{\mathbf{c} \in \mathcal{C}} \tilde{\mathbf{b}}^\top \mathbf{c}$ ;  $\text{step} \leftarrow \tau_d/3$ ;  $(\mathbf{z}^*, v^*) \leftarrow (\mathbf{z}, \tilde{\mathbf{b}}^\top \mathbf{z})$  ▷ initialize
4: for  $r = 1, \dots, R$  ( $R=12$  refine steps) do
5:    $\mathbf{z}^+ \leftarrow \text{PROJECTTOM}(\mathbf{z} + \text{step} \cdot u)$ 
6:   if  $\tilde{\mathbf{b}}^\top \mathbf{z}^+ > v^*$  then  $(\mathbf{z}^*, v^*, \mathbf{z}) \leftarrow (\mathbf{z}^+, \tilde{\mathbf{b}}^\top \mathbf{z}^+, \mathbf{z}^+)$  ▷ accept, recenter
7:   else  $\text{step} \leftarrow \text{step}/2$  ▷ backtrack
8:   end if
9: end for
10: return  $\mathbf{z}^*$ 



---


11: function PROJECTTOM( $\mathbf{z}$ ) ▷ approx. projection onto  $\mathcal{M} = \{d_k \leq \tau_d\}$ 
12:   if  $d_k(\mathbf{z}) \leq \tau_d$  then return  $\mathbf{z}$ 
13:   end if
14:    $\text{nn} \leftarrow k$  nearest anchors to  $\mathbf{z}$ ;  $\mathbf{z}^{\text{tgt}} \leftarrow \text{mean}(\{\mathbf{z}_i : i \in \text{nn}\})$ 
15:   bisect  $\alpha \in [0, 1]$  for the smallest  $\alpha$  with  $d_k(\mathbf{z} + \alpha(\mathbf{z}^{\text{tgt}} - \mathbf{z})) \leq \tau_d$ 
16:   return  $\mathbf{z} + \alpha(\mathbf{z}^{\text{tgt}} - \mathbf{z})$ 
17: end function

```

B Computational results: cumulative-regret behavior

We test the optimization/regret behavior of CM-TS in a *synthetic* simulator (NumPy, no image generation): the action set is the $M=228$ kNN-validity set $\mathcal{M}$ in PCA space, labels are drawn from a known logistic β^* against a fixed competitor, and the acquisition is **Algorithm 1'** (the continuous approximate oracle of Appendix A; per-round boundary maximizers + projected-gradient refinement onto $\partial\mathcal{M}$ — not a fixed pool, so this is genuinely the continuous algorithm, not a finite-arm bandit), *not* the exact Algorithm 1. Horizon $T=200$, small warm-start $n_0=6$ (initial designs), tuned exploration $v=0.4$ (tuning analysis below); baselines are greedy ($v=0$, pure exploitation) and random-valid. This isolates regret behavior; it is *not* a test of representation realism (that is the field study, journal §Validation). Note the simulation exercises *none* of the three idealizations the theorem assumes (exact SOCP / g_t -projection / worst-case v_T): it uses the continuous approximate oracle (not the exact SOCP), a Euclidean clip (not the g_t -projection), and $v \approx 0.4$ (not the worst-case $v_T = \Theta(\sqrt{d} \log T)$). Appendix C is therefore a qualitative no-regret/robustness check on a related procedure, not a validation of the bound. The synthetic simulator uses the equivalent *difference* encoding $\mathbf{z} - a$ (a d -dimensional β , no free intercept) rather than the body's augmented $(1, \mathbf{z})$; both encode the same beat-the-competitor problem — same $\mathbf{z}^*$, same regret target, both $\tilde{O}(d^{3/2}\sqrt{T})$ (App. C encoding note). Every figure and number below is from *one* run with a single replication count (40 replications throughout), reproducible from `make_paper_figs.py` under Python 3.13.7 / NumPy 2.4.3 with fixed RNG seeds 0–39 (which prints the `results` dict the captions quote; the `default_rng(seed)` stream is NumPy-version-sensitive, so the exact values pin to this environment). Two instance families: *easy* (well-specified, isotropic vocabulary) and *hard* (ill-conditioned: the decision-relevant direction has small spread, so it is under-explored).

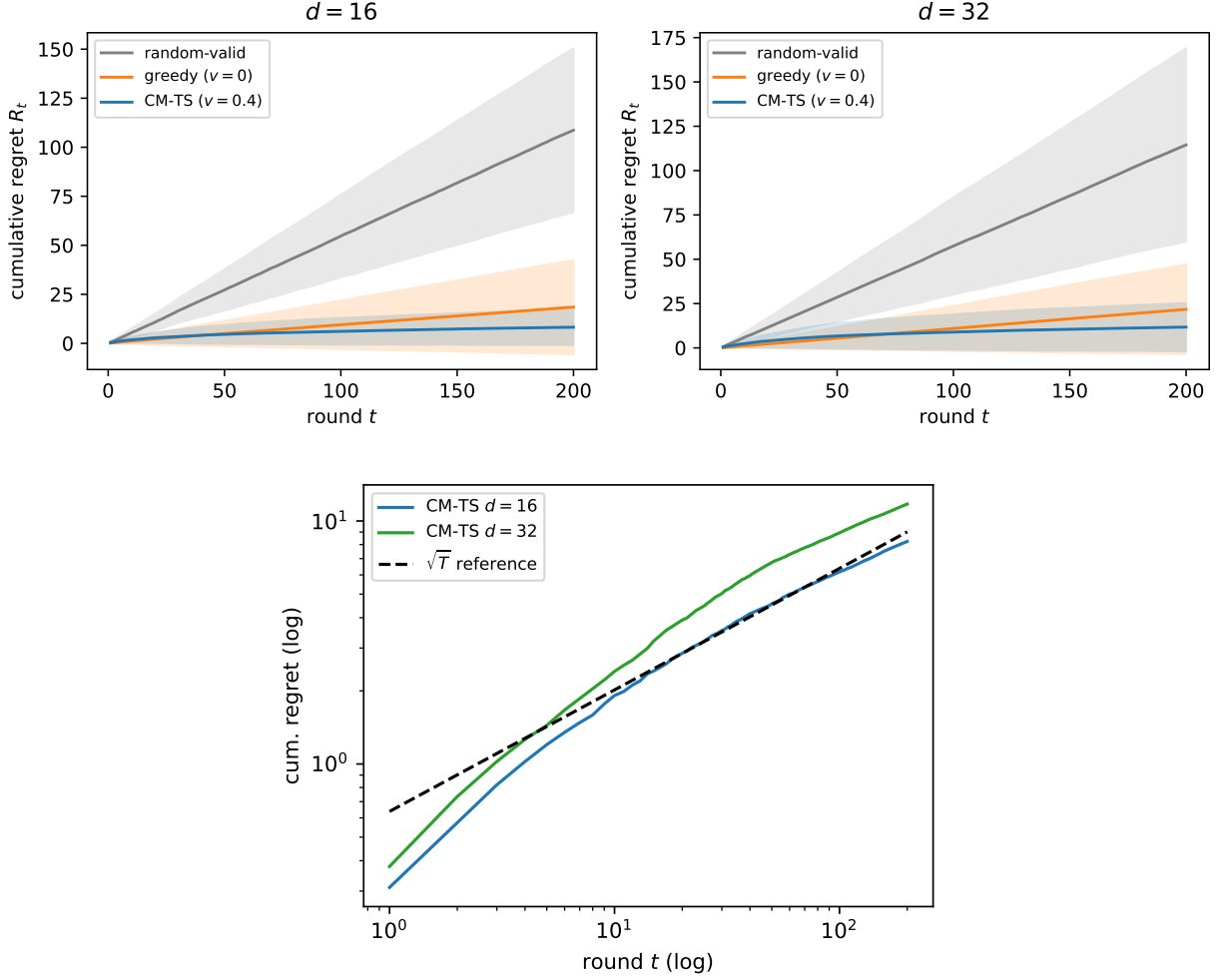


Figure 1: **Top:** cumulative beat-the-competitor regret R_t vs round (mean ± 1 std band over 40 replications), CM-TS ($v=0.4$) vs greedy ($v=0$) vs random-valid, $d=16$ (left) and $d=32$ (right). Tuned CM-TS beats random-valid by ~ 9.8 – $13\times$ and beats greedy by $\sim 2\times$ in mean (8.2 vs 18.5 at $d=16$; 11.7 vs 21.7 at $d=32$); note greedy’s much wider band — its catastrophic-tail replications (Fig. 3). **Bottom:** log-log R_t with a $\sqrt{T}$ reference. Fitted slopes in T are 0.43 ($d=16$) and 0.40 ($d=32$): sublinear and consistent with the $\tilde{O}(\sqrt{T})$ rate of Theorem 1 (we do not claim the figure *confirms* the exponent).

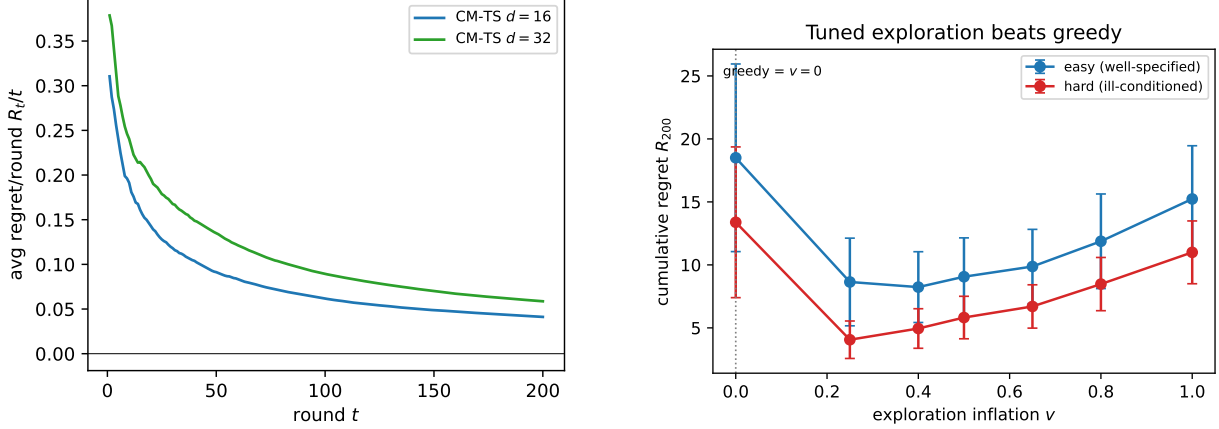


Figure 2: **Left:** average regret per round $R_t/t \rightarrow 0$ — the no-regret signature ($0.091 \rightarrow 0.041$ at $d=16$, $0.135 \rightarrow 0.059$ at $d=32$ over $T=200$). **Right (tuning):** cumulative R_T vs exploration inflation v ($v=0$ is exactly greedy). There is an interior optimum: greedy under-explores and tuned CM-TS beats it on *both* instance families — easy $18.5 \rightarrow 8.2$ (best $v=0.4$), hard $13.4 \rightarrow 4.1$ (best $v=0.25$); $v \gtrsim 1$ over-explores. The fixed default $v=0.4$ used elsewhere (and in the table below, hard CM-TS $R_T=5.0$) is near this optimum and well below the conservative worst-case $v_T = \sqrt{d \log T}$. All panels are the *same* 40-replication run, so the greedy point here equals the table’s greedy mean.

Why CM-TS and not greedy: the robustness story. On most replications greedy is fine — its *median* regret is excellent. But with no exploration guarantee it *gets stuck* on a fraction of replications (commits to a suboptimal design and never collects the data to correct), producing a heavy tail; the tuning sweep (Fig. 2, right) shows the interior optimum. Tuned CM-TS trades a little median performance for a bounded tail. On the hard instance ($d=16$, 40 replications):

hard instance, $T=200$ (40-replication run)	mean R_T	90th pct	max
greedy ($v=0$)	13.4	47.5	64.3
CM-TS ($v=0.4$)	4.95	10.4	24.9

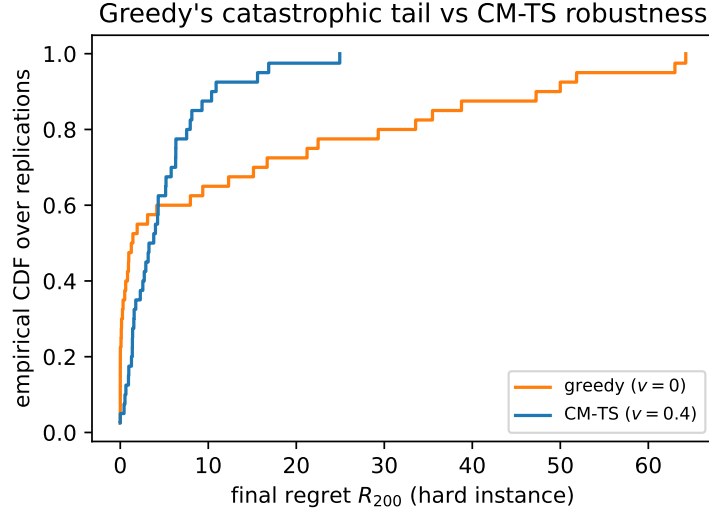


Figure 3: Empirical CDF of final regret R_T over replications (hard instance, 40 replications). Greedy’s distribution has a long right tail (worst replication $R_T=64.3$, 90th pct 47.5); CM-TS concentrates (max 24.9, 90th pct 10.4).

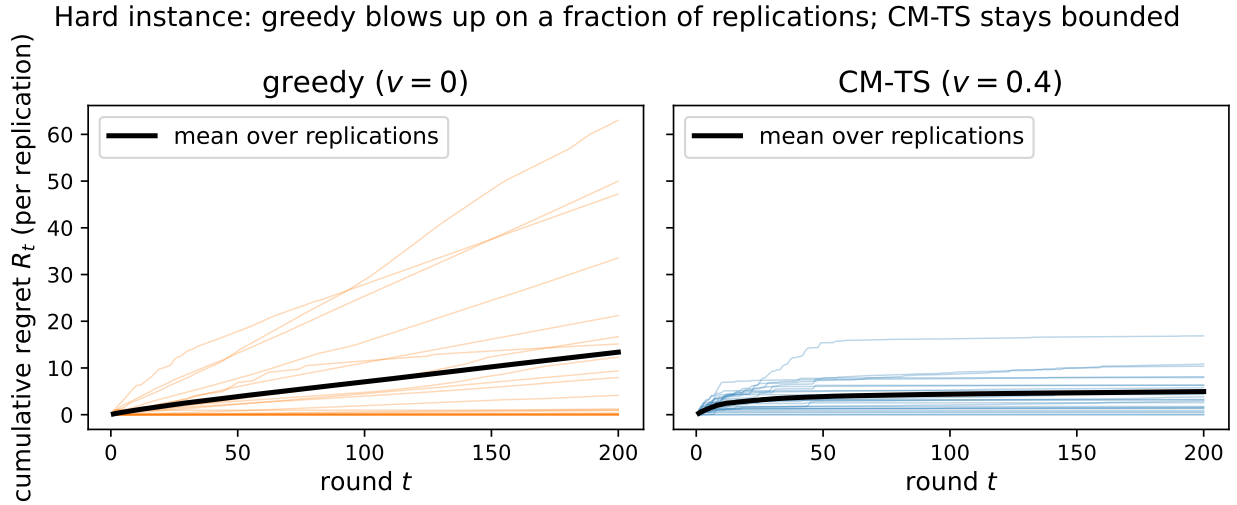


Figure 4: Per-replication cumulative-regret paths R_t on the hard instance ($d=16$; 25 of the 40 replications shown, black = mean). A fraction of *greedy* replications blow up (linear-looking growth — greedy commits early and gets stuck), while every *CM-TS* path stays bounded. This illustrates the empirical robustness motivation for exploration: greedy can get stuck, while CM-TS reduces catastrophic-tail outcomes in this simulation.

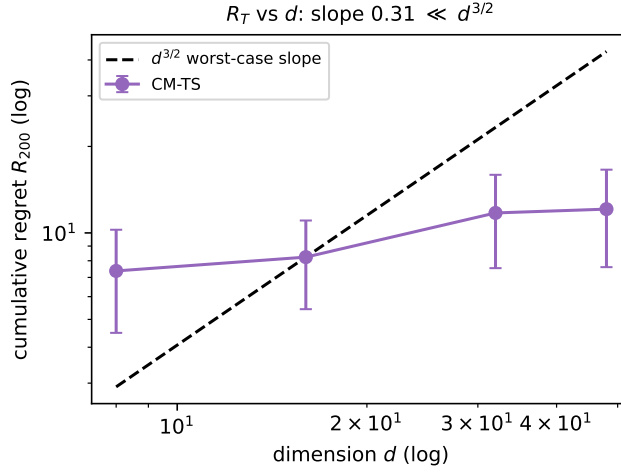


Figure 5: Cumulative R_T vs dimension $d \in \{8, 16, 32, 48\}$ (log-log, CM-TS, 40 replications; the $d=16, 32$ points equal the main-figure means). Regret rises with d but *well below* the $d^{3/2}$ worst-case slope (empirical slope ≈ 0.31 , distinct from the T -slopes 0.40 – 0.43 above): the bound is a non-tight upper bound on these well-specified instances, as expected. We report this honestly rather than as a validation of the $d^{3/2}$ exponent.

Summary. On a genuinely continuous (non-pool) action set, CM-TS is no-regret ($R_t/t \rightarrow 0$, Fig. 2 left), grows sublinearly in T (consistent with $\tilde{O}(\sqrt{T})$, Fig. 1), rises only mildly with d (Fig. 5), and with tuned exploration $v \approx 0.4$ beats both greedy and random in *mean* and — the load-bearing point — in the *tail* (Fig. 3): greedy’s lack of a guarantee shows up as catastrophic stuck-replications that tuned CM-TS eliminates. The simulation tests the optimization/regret behavior on a well-specified instance; it is consistent with, but does not by itself prove, the formal rate.

C CM-TS regret proof

We prove the optimization-side guarantee for CM-TS. The setting is the augmented feature $\phi(\mathbf{z}) = (1, \mathbf{z}^\top)^\top \in \mathbb{R}^{d+1}$, so $\Pr(y_t=1 \mid \mathbf{z}_t) = \sigma(\boldsymbol{\beta}^{\star\top} \phi(\mathbf{z}_t))$ (the body's $\sigma(\beta_0 + \boldsymbol{\beta}^\top \mathbf{z})$, competitor in β_0 ; this augmented encoding replaces the difference encoding $\mathbf{z} - a$ of `current_proof.md`, both valid, $d+1$ vs d , with κ_σ at logit radius $R_\phi S$ here vs $2RS$ there); $\|\phi(\mathbf{z})\| \leq R_\phi := \sqrt{1 + R^2}$, $\|\boldsymbol{\beta}^*\| \leq S$. The optimum $\mathbf{z}^* = \arg \max_{\mathbf{z} \in \mathcal{M}} \boldsymbol{\beta}^{\star\top} \mathbf{z}$ may lie on $\partial\mathcal{M}$; $\mathcal{M}$ may be non-convex. The proof uses (A0) the realizable, predictable logistic observation model (stated in Theorem 1), (A1) $\|\boldsymbol{\beta}^*\| \leq S$, (A2) $\mathcal{M}$ compact with $\sup_{\mathcal{M}} \|\mathbf{z}\| \leq R$, (A3) ridge $\lambda = \sigma_{\min}^{-2}$, and the Abbasi-Yadkori self-normalized martingale inequality [Abbasi-Yadkori et al., 2011].

Theorem 1 (Regret of CM-TS). *Run CM-TS (variance-inflated draw $v_T = \beta_T(1/T^2)$; exact global $\arg \max$ over $\mathcal{M}$; g_t -projected ridge estimator) against the fixed competitor. Assume (A0) the realizable, predictable observation model: conditional on $\mathcal{F}_{t-1}$, the design $\mathbf{z}_t$ is $\mathcal{F}_{t-1}$ -predictable (chosen before y_t) and $y_t \sim \text{Bernoulli}(\sigma(\boldsymbol{\beta}^{\star\top} \phi(\mathbf{z}_t)))$ (the true preference is logistic in ϕ), together with (A1) $\|\boldsymbol{\beta}^*\| \leq S$, (A2) $\mathcal{M}$ compact, $\sup_{\mathcal{M}} \|\mathbf{z}\| \leq R$, (A3) ridge $\lambda = \sigma_{\min}^{-2}$. Then the cumulative beat-the-competitor regret $R_T = \sum_{t=1}^T [\sigma(\boldsymbol{\beta}^{\star\top} \phi(\mathbf{z}^*)) - \sigma(\boldsymbol{\beta}^{\star\top} \phi(\mathbf{z}_t))]$ satisfies*

$$\mathbb{E}[R_T] = \tilde{O}(d^{3/2}\sqrt{T}) = O\left(\frac{1}{p_{\text{opt}}\kappa_\sigma^{3/2}}d^{3/2}\sqrt{T}(\log T)^{3/2}\right), \quad p_{\text{opt}} = 1 - \Phi(1), \quad \kappa_\sigma = \sigma(R_\phi S)(1 - \sigma(R_\phi S)).$$

An inexact acquisition with per-round error $\epsilon_t \geq 0$ adds $\frac{1}{4} \sum_t \epsilon_t$.

Scope of the constant. The bound carries $\kappa_\sigma^{-3/2}$ with $\kappa_\sigma = \sigma(R_\phi S)(1 - \sigma(R_\phi S))$, which is exponentially small when the logit range $R_\phi S$ is large; the guarantee is therefore meaningful in the bounded-logit regime $R_\phi S = O(1)$. The norm projection $\{\|\boldsymbol{\beta}\| \leq S\}$ is precisely what caps the logit range at this $O(1)$, T -independent scale so that κ_σ stays a positive constant — the standard GLM-bandit device [Filippi et al., 2010]; some such cap is necessary, since ridge regularization alone bounds the unconstrained estimator only by $\|\tilde{\boldsymbol{\beta}}_t\| = O(tR_\phi/\lambda)$ (growing with t), which would drive $\kappa_\sigma \approx e^{-\Theta(T)}$. One may take S to be any known upper bound on $\|\boldsymbol{\beta}^*\|$ (e.g. a few prior standard deviations, or domain knowledge of the preference-vector scale); the smallest safe S minimizes the $\kappa_\sigma^{-3/2}$ constant. Relocating the $1/\kappa_\sigma$ dependence to a lower-order term (the self-concordant route of Faury et al., 2020, which still assumes $\|\boldsymbol{\beta}^*\| \leq S$) is a known refinement we do not pursue here.

What the theorem covers (positioning). The guarantee is for an *idealized* CM-TS variant with *exact* acquisition (Algorithm 1) and *mean-map* (g_t) *projection*. The deployed implementation uses two computational relaxations — *approximate continuous acquisition* (Algorithm 1') and *Euclidean clipping* of the estimator — and the empirical study (App. B) evaluates the *relaxed* algorithm. Theorem 1 is not claimed to cover the deployed algorithm: the acquisition gap enters additively as $\frac{1}{4} \sum_t \epsilon_t$ (controlled only if that cumulative gap is), and the Euclidean clip is an estimator relaxation *not* covered by the bound (§3). *Misspecification.* Under model misspecification the theorem should be read as applying to the well-specified projection target (the logistic-in- ϕ preference); empirical robustness under misspecification is assessed separately.

Setup. $\mathcal{F}_{t-1} := \sigma(\mathbf{z}_1, y_1, \dots, \mathbf{z}_{t-1}, y_{t-1})$, $\lambda = \sigma_{\min}^{-2}$, $V_t := \lambda I + \sum_{s < t} \phi(\mathbf{z}_s)\phi(\mathbf{z}_s)^\top$, and $H_{t-1} := \lambda I + \sum_{s < t} \dot{\sigma}(\hat{\boldsymbol{\beta}}_{t-1}^\top \phi(\mathbf{z}_s))\phi(\mathbf{z}_s)\phi(\mathbf{z}_s)^\top$ (the analyzed Laplace posterior covariance is $\boldsymbol{\Sigma}_{t-1} = H_{t-1}^{-1}$, the generic $\boldsymbol{\Sigma}_{t-1}$ of Algorithm 2). The estimator is the unconstrained ridge quasi-MLE $\tilde{\boldsymbol{\beta}}_{t-1}$ projected onto $C := \{\|\boldsymbol{\beta}\| \leq S\} \ni \boldsymbol{\beta}^*$ in the mean-map norm: $\hat{\boldsymbol{\beta}}_{t-1} = \arg \min_{\boldsymbol{\beta} \in C} \|g_t(\boldsymbol{\beta}) - g_t(\tilde{\boldsymbol{\beta}}_{t-1})\|_{V_t^{-1}}$, $g_t(\boldsymbol{\beta}) = \lambda\boldsymbol{\beta} + \sum_{s < t} \sigma(\boldsymbol{\beta}^\top \phi(\mathbf{z}_s))\phi(\mathbf{z}_s)$ (not Euclidean). Then $\|\hat{\boldsymbol{\beta}}_{t-1}\| \leq S$, so $|\hat{\boldsymbol{\beta}}_{t-1}^\top \phi(\mathbf{z}_s)| \leq R_\phi S$

and $\dot{\sigma} \geq \kappa_\sigma := \sigma(R_\phi S)(1 - \sigma(R_\phi S))$, giving the deterministic sandwich:

$$\kappa_\sigma V_t \preceq H_{t-1} \preceq V_t, \quad \|v\|_{V_t^{-1}} \leq \|v\|_{H_{t-1}^{-1}}, \quad \|v\|_{V_t} \leq \kappa_\sigma^{-1/2} \|v\|_{H_{t-1}}. \quad (\text{S})$$

The support function $J(\theta) = \max_{\mathbf{z} \in \mathcal{M}} \theta^\top \phi(\mathbf{z})$ is convex, R_ϕ -Lipschitz, with maximizer $\mathbf{z}(\theta) \in \mathcal{M}$ (compactness), $\phi(\mathbf{z}(\theta)) \in \partial J(\theta)$; $\mathbf{z}_t = \mathbf{z}(\tilde{\beta}_t)$, $J(\tilde{\beta}_t) = \tilde{\beta}_t^\top \phi(\mathbf{z}_t)$. *No convexity of $\mathcal{M}$ is used.*

Lemma 1 (self-normalized concentration). *With prob. $\geq 1 - \delta$, for all t , $\|\hat{\beta}_{t-1} - \beta^*\|_{V_t} \leq \beta_t(\delta) := \frac{2}{\kappa_\sigma} (\sqrt{(d+1) \log(\frac{1+tR_\phi^2/\lambda}{\delta})} + \sqrt{\lambda}S)$. Proof (3 steps). (i) Ridge stationarity of $\tilde{\beta}_{t-1}$ gives $g_t(\tilde{\beta}_{t-1}) = \sum_{s < t} y_s \phi(\mathbf{z}_s)$, so $g_t(\tilde{\beta}_{t-1}) - g_t(\beta^*) = S_t - \lambda \beta^*$ with $S_t = \sum_{s < t} (y_s - \sigma(\beta^{*\top} \phi(\mathbf{z}_s))) \phi(\mathbf{z}_s)$. By (A0) the increments are a $\mathcal{F}_s$ -martingale-difference sequence with $y_s - \sigma(\beta^{*\top} \phi(\mathbf{z}_s)) \in [-1, 1]$, hence $\frac{1}{2}$ -sub-Gaussian (range 1, Hoeffding; the Abbasi-Yadkori bound below uses the conservative noise scale 1, with no separate symbol — R is reserved for the feature-norm bound), $\|\phi\| \leq R_\phi$; a loosened form of the Abbasi-Yadkori bound [Abbasi-Yadkori et al., 2011] gives $\|S_t\|_{V_t^{-1}} \leq \sqrt{(d+1) \log(\frac{1+tR_\phi^2/\lambda}{\delta})}$ (dropping the factor-2 and the $1/(d+1)$ inside the log for a cleaner constant; this only inflates the bound), hence $\|g_t(\tilde{\beta}_{t-1}) - g_t(\beta^*)\|_{V_t^{-1}} \leq \sqrt{(d+1) \log(\cdot)} + \sqrt{\lambda}S$. (ii) $\beta^* \in C$ and $\hat{\beta}_{t-1}$ minimizes the mean-map distance over C , so by the triangle inequality $\|g_t(\hat{\beta}_{t-1}) - g_t(\beta^*)\|_{V_t^{-1}} \leq 2(\sqrt{(d+1) \log(\cdot)} + \sqrt{\lambda}S)$ (this contraction is derived inline here and requires the g_t -norm projection, not Euclidean; Filippi et al., 2010 is cited for the mean-map projection concept, not a precise lemma number). (iii) $g_t(\hat{\beta}_{t-1}) - g_t(\beta^*) = \bar{G}_t(\hat{\beta}_{t-1} - \beta^*)$, $\bar{G}_t \succeq \kappa_\sigma V_t$ (MVT, both in C); writing $A = V_t^{1/2}$, $B = A^{-1} \bar{G}_t A^{-1} \succeq \kappa_\sigma I$, $\|\bar{G}_t u\|_{V_t^{-1}}^2 = (Au)^\top B^2 (Au) \geq \kappa_\sigma^2 \|u\|_{V_t}^2$, so $\|\hat{\beta}_{t-1} - \beta^*\|_{V_t} \leq \kappa_\sigma^{-1} \|g_t(\hat{\beta}_{t-1}) - g_t(\beta^*)\|_{V_t^{-1}} = \beta_t(\delta)$. $\square$*

Write $\gamma_t = \beta_t(1/T)$, $\mathcal{A}_{t-1} = \{\|\hat{\beta}_{t-1} - \beta^*\|_{V_t} \leq \gamma_t\} \in \mathcal{F}_{t-1}$ (holds for all t w.p. $\geq 1 - 1/T$). Set $v_T = \beta_T(1/T^2) \geq \gamma_t \forall t \leq T$. With $\rho_T = 3\sqrt{(d+1) \log T}$ (so for $T \geq 2$, $\Pr(\|\eta_t\| > \rho_T \mid \mathcal{F}_{t-1}) \leq T^{-2}$ by the Gaussian-norm / χ^2 tail of Laurent and Massart, 2000, $\eta_t \sim \mathcal{N}(0, I_{d+1})$, since $3\sqrt{(d+1) \log T} \geq \sqrt{d+1} + \sqrt{4 \log T}$): on $\{\|\eta_t\| \leq \rho_T\}$, by (S), $\|\tilde{\beta}_t - \hat{\beta}_{t-1}\|_{V_t} \leq \kappa_\sigma^{-1/2} v_T \rho_T =: \gamma_t^{\text{TS}}$, $w_t := \gamma_t + \gamma_t^{\text{TS}}$.

Lemma 2 (constant optimism). *On $\mathcal{A}_{t-1}$, $\Pr(J(\tilde{\beta}_t) \geq J(\beta^*) \mid \mathcal{F}_{t-1}) \geq p_{\text{opt}} = 1 - \Phi(1)$. Since $\mathbf{z}^* \in \mathcal{M}$, $J(\tilde{\beta}_t) \geq \tilde{\beta}_t^\top \phi(\mathbf{z}^*) \sim \mathcal{N}(\tilde{\beta}_t^\top \phi(\mathbf{z}^*), v_T^2 \|\phi(\mathbf{z}^*)\|_{H_{t-1}^{-1}}^2)$, so the probability is $\Phi^c((\beta^* - \hat{\beta}_{t-1})^\top \phi(\mathbf{z}^*) / (v_T \|\phi(\mathbf{z}^*)\|_{H_{t-1}^{-1}}))$. On $\mathcal{A}_{t-1}$ the numerator is $\leq \gamma_t \|\phi(\mathbf{z}^*)\|_{V_t^{-1}}$ and $\|\phi(\mathbf{z}^*)\|_{H_{t-1}^{-1}} \geq \|\phi(\mathbf{z}^*)\|_{V_t^{-1}}$ by (S) (only $H_{t-1} \preceq V_t$, no κ_σ), so the argument is $\leq \gamma_t / v_T \leq 1$ and $\Phi^c(\leq 1) \geq p_{\text{opt}}$. Only $\mathbf{z}^* \in \mathcal{M}$ and a scalar Gaussian are used — valid on $\partial \mathcal{M}$, non-convex $\mathcal{M}$. $\square$*

Regret decomposition + Lemma 3. With $\sigma \frac{1}{4}$ -Lipschitz, $r_t \leq \frac{1}{4} \Delta_t$. *Inexact-oracle convention (P-PROOF2):* let $\epsilon_t := J(\tilde{\beta}_t) - \tilde{\beta}_t^\top \phi(\mathbf{z}_t) \geq 0$ be the per-round acquisition suboptimality ($\epsilon_t \equiv 0$ for the exact oracle of Algorithm 1; $\epsilon_t \geq 0$ for Algorithm 1'). With $J(\beta^*) = \beta^{*\top} \phi(\mathbf{z}^*)$,

$$\Delta_t := \beta^{*\top} \phi(\mathbf{z}^*) - \beta^{*\top} \phi(\mathbf{z}_t) = \underbrace{[J(\beta^*) - J(\tilde{\beta}_t)]}_{\Delta_t^{\text{TS}}} + \underbrace{[J(\tilde{\beta}_t) - \tilde{\beta}_t^\top \phi(\mathbf{z}_t)]}_{=\epsilon_t} + \underbrace{(\tilde{\beta}_t - \beta^*)^\top \phi(\mathbf{z}_t)}_{\Delta_t^{\text{est}}},$$

so $\Delta_t = \Delta_t^{\text{TS}} + \epsilon_t + \Delta_t^{\text{est}}$; the extra ϵ_t enters *before* the $\frac{1}{4}$ -Lipschitz multiply, giving $+\frac{1}{4} \sum_t \epsilon_t$. (The intercept cancels in Δ_t : the first coordinate of $\phi(\mathbf{z}^*) - \phi(\mathbf{z}_t)$ is 0, so Δ_t is design-only.) For the exact oracle $\epsilon_t = 0$ and $\Delta_t = \Delta_t^{\text{TS}} + \Delta_t^{\text{est}}$; we bound these two. *Tail constant.* Since $\tilde{\beta}_t - \hat{\beta}_{t-1} = v_T H_{t-1}^{-1/2} \eta_t$, by (S) $\|\tilde{\beta}_t - \hat{\beta}_{t-1}\|_{V_t} \leq \kappa_\sigma^{-1/2} v_T \|\eta_t\|$, and $\|\phi(\mathbf{z}_t)\|_{V_t^{-1}} \leq R_\phi / \sqrt{\lambda}$ (as $V_t \succeq \lambda I$); Cauchy-Schwarz with $\mathbb{E}\|\eta_t\|^2 = d+1$ and $\Pr(\|\eta_t\| > \rho_T \mid \mathcal{F}_{t-1}) \leq T^{-2}$ gives

$$\mathbb{E}[\|\eta_t\| \mathbf{1}\{\|\eta_t\| > \rho_T\} \mid \mathcal{F}_{t-1}] \leq \sqrt{\mathbb{E}\|\eta_t\|^2 \cdot \Pr(\|\eta_t\| > \rho_T)} \leq \sqrt{(d+1)T^{-2}} = \frac{\sqrt{d+1}}{T},$$

so each tail contributes at most $\zeta_t := \frac{R_\phi}{\sqrt{\lambda}} \kappa_\sigma^{-1/2} v_T \frac{\sqrt{d+1}}{T}$ ($\sum_t \zeta_t = \tilde{O}(d)$). Splitting on $\{\|\eta_t\| \leq \rho_T\}$ (main: $\|\tilde{\beta}_t - \beta^*\|_{V_t} \leq w_t$) and bounding the tail by $2\zeta_t$ (the γ_t and $\kappa_\sigma^{-1/2} v_T \|\eta_t\|$ pieces), $\mathbb{E}[\Delta_t^{\text{est}} | \mathcal{F}_{t-1}] \leq w_t \mathbb{E}[\|\phi(\mathbf{z}_t)\|_{V_t^{-1}} | \mathcal{F}_{t-1}] + 2\zeta_t$. **Lemma 3 (optimism reduction, the Thompson-sampling resample coupling of Abeille and Lazaric, 2017, Agrawal and Goyal, 2013).** On $\mathcal{A}_{t-1}$, $\mathbb{E}[\Delta_t^{\text{TS}} | \mathcal{F}_{t-1}] \leq \frac{\gamma_t^{\text{TS}}}{p_{\text{opt}}} \mathbb{E}[\|\phi(\mathbf{z}_t)\|_{V_t^{-1}} | \mathcal{F}_{t-1}] + \frac{\zeta_t}{p_{\text{opt}}}$, where $\Delta_t^{\text{TS}} = J(\beta^*) - J(\tilde{\beta}_t)$.

The entire argument uses only the support function $J(\theta) = \max_{\mathbf{z} \in \mathcal{M}} \theta^\top \phi(\mathbf{z})$. The kNN-validity set $\mathcal{M}$ enters at exactly two points, both through J alone and never through convexity, smoothness, or connectedness of $\mathcal{M}$: (i) J is finite, convex and R_ϕ -Lipschitz with its max attained — purely from compactness of $\mathcal{M}$ (a finite union of closed, bounded ball-cap intersections, hence compact) together with J being a pointwise max of linear functions of θ ; and (ii) the feasibility inequality of Step 5. $\mathcal{M}$ is nonconvex, possibly nonsmooth and disconnected; no property of $\mathcal{M}$ beyond compactness and the definition of J is used. Statements are conditional on $\mathcal{F}_{t-1}$; only Step 2 uses $\mathcal{A}_{t-1} \in \mathcal{F}_{t-1}$ (via Lemma 2), the rest hold unconditionally given $\mathcal{F}_{t-1}$.

Step 0 (objects, attainment, measurability). As ϕ is continuous and $\mathcal{M}$ compact, $\phi(\mathcal{M})$ is compact, so J is finite, convex, R_ϕ -Lipschitz with the max attained; $\mathbf{z}(\theta) \in \arg \max_{\mathbf{z} \in \mathcal{M}} \theta^\top \phi(\mathbf{z})$ is a measurable maximizer selection (measurable-maximum theorem; $\mathcal{M}$ a compact metric set and $\theta^\top \phi$ continuous), so $\phi(\mathbf{z}(\theta))$ is a well-defined $\mathcal{F}_{t-1}$ -measurable random element with $\phi(\mathbf{z}(\theta)) \in \partial J(\theta)$. The played action is $\mathbf{z}_t = \mathbf{z}(\tilde{\beta}_t)$, $J(\tilde{\beta}_t) = \tilde{\beta}_t^\top \phi(\mathbf{z}_t)$ (exact oracle; the inexact gap ϵ_t is already separated in the decomposition). Here $\mathbf{z}^* = \mathbf{z}(\beta^*)$ is deterministic given $\mathcal{F}_{t-1}$, and $\beta^*, \tilde{\beta}_{t-1}, H_{t-1}, V_t$ are $\mathcal{F}_{t-1}$ -measurable.

Step 1 (resample sharing the conditional law). Let $\tilde{\beta}'_t$ have the same $\mathcal{F}_{t-1}$ -conditional law as $\tilde{\beta}_t$, namely $\mathcal{N}(\hat{\beta}_{t-1}, v_T^2 H_{t-1}^{-1})$ (an independent copy is one construction; only equality in conditional law is used — independence is not needed, which avoids any joint-measurability assumption), and set $\mathbf{z}'_t = \mathbf{z}(\tilde{\beta}'_t)$. Then for any measurable $h \geq 0$, $\mathbb{E}[h(\tilde{\beta}'_t) | \mathcal{F}_{t-1}] = \mathbb{E}[h(\tilde{\beta}_t) | \mathcal{F}_{t-1}]$; in particular $\mathbb{E}[\|\phi(\mathbf{z}'_t)\|_{V_t^{-1}} | \mathcal{F}_{t-1}] = \mathbb{E}[\|\phi(\mathbf{z}_t)\|_{V_t^{-1}} | \mathcal{F}_{t-1}]$ (equal conditional law $\Rightarrow$ equal conditional expectation of any nonneg functional).

Step 2 (optimism lower-bounds $J(\beta^)$).* Let $E_t^{\text{opt}} := \{J(\tilde{\beta}'_t) \geq J(\beta^*)\}$. By Lemma 2 applied to $\tilde{\beta}'_t$ (same law), on $\mathcal{A}_{t-1}$, $\Pr(E_t^{\text{opt}} | \mathcal{F}_{t-1}) \geq p_{\text{opt}}$. On E_t^{opt} , $J(\beta^*) \leq J(\tilde{\beta}'_t)$ pointwise; as $J(\beta^*)$ is $\mathcal{F}_{t-1}$ -measurable, $J(\beta^*) \leq \mathbb{E}[J(\tilde{\beta}'_t) | E_t^{\text{opt}}, \mathcal{F}_{t-1}]$ (monotonicity of conditional expectation).

Step 3 (Thompson term via the two J 's). $\mathbb{E}[\Delta_t^{\text{TS}} | \mathcal{F}_{t-1}] = J(\beta^*) - \mathbb{E}[J(\tilde{\beta}_t) | \mathcal{F}_{t-1}] \leq \mathbb{E}[J(\tilde{\beta}'_t) | E_t^{\text{opt}}, \mathcal{F}_{t-1}] - \mathbb{E}[J(\tilde{\beta}_t) | \mathcal{F}_{t-1}]$ (substitute Step 2; $J(\beta^*)$ is an $\mathcal{F}_{t-1}$ -constant).

Step 4 (Jensen lower bound on $\mathbb{E}[J(\tilde{\beta}_t)]$). J convex (Step 0) $\Rightarrow$ conditional Jensen: $\mathbb{E}[J(\tilde{\beta}_t) | \mathcal{F}_{t-1}] \geq J(\mathbb{E}[\tilde{\beta}_t | \mathcal{F}_{t-1}]) = J(\hat{\beta}_{t-1})$ (the draw mean is $\hat{\beta}_{t-1}$; integrability from R_ϕ -Lipschitz J and the Gaussian draw). Substituting, $\mathbb{E}[\Delta_t^{\text{TS}} | \mathcal{F}_{t-1}] \leq \mathbb{E}[J(\tilde{\beta}'_t) - J(\hat{\beta}_{t-1}) | E_t^{\text{opt}}, \mathcal{F}_{t-1}]$ ($J(\hat{\beta}_{t-1})$ is an $\mathcal{F}_{t-1}$ -constant, moved inside).

Step 5 (support-function / subgradient bound — the only place $\mathcal{M}$ enters). For any θ , $J(\theta) = \theta^\top \phi(\mathbf{z}(\theta))$ while $J(\hat{\beta}_{t-1}) \geq \hat{\beta}_{t-1}^\top \phi(\mathbf{z}(\theta))$ because $\mathbf{z}(\theta) \in \mathcal{M}$ is feasible in the max defining $J(\hat{\beta}_{t-1})$; subtracting, $J(\theta) - J(\hat{\beta}_{t-1}) \leq (\theta - \hat{\beta}_{t-1})^\top \phi(\mathbf{z}(\theta)) \leq \|\theta - \hat{\beta}_{t-1}\|_{V_t} \|\phi(\mathbf{z}(\theta))\|_{V_t^{-1}}$ (dual-norm Cauchy-Schwarz). At $\theta = \tilde{\beta}'_t$: $J(\tilde{\beta}'_t) - J(\hat{\beta}_{t-1}) \leq D' P'$ with $D' := \|\tilde{\beta}'_t - \hat{\beta}_{t-1}\|_{V_t}$, $P' := \|\phi(\mathbf{z}'_t)\|_{V_t^{-1}}$. This uses only the support-function definition, feasibility of $\mathbf{z}(\theta)$, and Cauchy-Schwarz — no convexity or geometry of $\mathcal{M}$.

(Conditioning, Steps 6–8.) All conditional expectations below additionally condition on $\mathcal{F}_{t-1}$ unless displayed otherwise. *Step 6 (split D' into the typical bound γ_t^{TS} and the Gaussian tail ζ_t).* Write $\tilde{\beta}'_t - \hat{\beta}_{t-1} = v_T H_{t-1}^{-1/2} \eta'_t$, $\eta'_t \sim \mathcal{N}(0, I_{d+1})$; by (S), $D' \leq \kappa_\sigma^{-1/2} v_T \|\eta'_t\|$ and $P' \leq R_\phi / \sqrt{\lambda}$ (as $V_t \succeq \lambda I$).

On $\{\|\eta'_t\| \leq \rho_T\}$, $D' \leq \kappa_\sigma^{-1/2} v_T \rho_T = \gamma_t^{\text{TS}}$, whence

$$\mathbb{E}[D'P' \mid E_t^{\text{opt}}] \leq \gamma_t^{\text{TS}} \mathbb{E}[P' \mid E_t^{\text{opt}}] + \frac{R_\phi}{\sqrt{\lambda}} \kappa_\sigma^{-1/2} v_T \mathbb{E}[\|\eta'_t\| \mathbf{1}\{\|\eta'_t\| > \rho_T\} \mid E_t^{\text{opt}}]$$

(main term: drop the ≤ 1 indicator since $\gamma_t^{\text{TS}} P' \geq 0$; tail term: $P' \leq R_\phi/\sqrt{\lambda}$). For the tail, the Step-7 transfer plus the Gaussian-tail bound *already displayed* above give $\mathbb{E}[\|\eta'_t\| \mathbf{1}\{\|\eta'_t\| > \rho_T\} \mid \mathcal{F}_{t-1}] \leq \sqrt{\mathbb{E}\|\eta'_t\|^2 \Pr(\|\eta'_t\| > \rho_T)} \leq \sqrt{(d+1)T^{-2}} = \frac{\sqrt{d+1}}{T}$ (Cauchy–Schwarz; $\mathbb{E}\|\eta'_t\|^2 = d+1$; Laurent–Massart $\Pr(\|\eta'_t\| > \rho_T) \leq T^{-2}$), so the tail term is $\leq \zeta_t/p_{\text{opt}}$ with $\zeta_t = \frac{R_\phi}{\sqrt{\lambda}} \kappa_\sigma^{-1/2} v_T \frac{\sqrt{d+1}}{T}$ (the same ζ_t as in the estimator term).

Step 7 (conditional-expectation transfer; remove the optimistic conditioning). For nonneg X with $\Pr(E_t^{\text{opt}} \mid \mathcal{F}_{t-1}) \geq p_{\text{opt}}$, $\mathbb{E}[X \mid E_t^{\text{opt}}, \mathcal{F}_{t-1}] = \mathbb{E}[X \mathbf{1}_{E_t^{\text{opt}}} \mid \mathcal{F}_{t-1}] / \Pr(E_t^{\text{opt}} \mid \mathcal{F}_{t-1}) \leq \mathbb{E}[X \mid \mathcal{F}_{t-1}] / p_{\text{opt}}$ (numerator: $X \mathbf{1} \leq X$; denominator $\geq p_{\text{opt}}$). With $X = P' = \|\phi(\mathbf{z}'_t)\|_{V_t^{-1}} \geq 0$ and then Step 1, $\mathbb{E}[P' \mid E_t^{\text{opt}}] \leq \mathbb{E}[P' \mid \mathcal{F}_{t-1}] / p_{\text{opt}} = \mathbb{E}[\|\phi(\mathbf{z}_t)\|_{V_t^{-1}} \mid \mathcal{F}_{t-1}] / p_{\text{opt}}$.

Step 8 (assemble). Chaining Steps 4 $\rightarrow$ 5 $\rightarrow$ 6 $\rightarrow$ 7,

$$\mathbb{E}[\Delta_t^{\text{TS}} \mid \mathcal{F}_{t-1}] \leq \mathbb{E}[D'P' \mid E_t^{\text{opt}}] \leq \gamma_t^{\text{TS}} \mathbb{E}[P' \mid E_t^{\text{opt}}] + \frac{\zeta_t}{p_{\text{opt}}} \leq \frac{\gamma_t^{\text{TS}}}{p_{\text{opt}}} \mathbb{E}[\|\phi(\mathbf{z}_t)\|_{V_t^{-1}} \mid \mathcal{F}_{t-1}] + \frac{\zeta_t}{p_{\text{opt}}}. \quad \square$$

Aggregation. Adding the two bounds, on $\mathcal{A}_{t-1}$: $\mathbb{E}[\Delta_t \mid \mathcal{F}_{t-1}] \leq (1 + \frac{1}{p_{\text{opt}}}) w_t \mathbb{E}[\|\phi(\mathbf{z}_t)\|_{V_t^{-1}} \mid \mathcal{F}_{t-1}] + (2 + \frac{1}{p_{\text{opt}}}) \zeta_t$. By the tower property ($\mathcal{A}_{t-1} \in \mathcal{F}_{t-1}$), dropping $\mathbf{1}_{\mathcal{A}_{t-1}} \leq 1$, Cauchy–Schwarz, and the elliptical-potential bound [Dani et al., 2008, Lem. 14] $\sum_t \|\phi(\mathbf{z}_t)\|_{V_t^{-1}}^2 \leq 2(d+1) \log(1 + \frac{TR_\phi^2}{(d+1)\lambda}) =: 2(d+1)L_T$ over the *played* features:

$$\sum_t \mathbb{E}[\Delta_t \mathbf{1}_{\mathcal{A}_{t-1}}] \leq (1 + \frac{1}{p_{\text{opt}}}) w_T \sqrt{2(d+1)L_T T} + \tilde{O}(d).$$

Since $\Delta_t \geq 0$ and $\mathcal{A}_{\leq T} \subseteq \mathcal{A}_{t-1}$, this bounds $\mathbb{E}[\sum_t \Delta_t \mathbf{1}_{\mathcal{A}_{\leq T}}]$; on $\mathcal{A}_{\leq T}^c$ (prob. $\leq 1/T$) each $\Delta_t \leq 2R_\phi S$, adding $O(R_\phi S)$. With $w_T = O(\kappa_\sigma^{-3/2} (d+1) \log T)$ and the $\frac{1}{4}$ factor, $\mathbb{E}[R_T] = \tilde{O}(d^{3/2} \sqrt{T})$, proving Theorem 1. The inexact-oracle term enters as derived in the decomposition above ($+\frac{1}{4} \sum_t \epsilon_t$, design-only). ■

References

- Y. Abbasi-Yadkori, D. Pál, and C. Szepesvári. Improved algorithms for linear stochastic bandits. In *NeurIPS*, 2011.
- M. Abeille and A. Lazaric. Linear thompson sampling revisited. In *AISTATS*, 2017.
- S. Agrawal and N. Goyal. Thompson sampling for contextual bandits with linear payoffs. In *ICML*, 2013.
- Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.
- V. Dani, T. P. Hayes, and S. M. Kakade. Stochastic linear optimization under bandit feedback. In *COLT*, 2008.
- L. Faury, M. Abeille, C. Calauzènes, and O. Fercoq. Improved optimistic algorithms for logistic bandits. In *ICML*, 2020.
- S. Filippi, O. Cappé, A. Garivier, and C. Szepesvári. Parametric bandits: The generalized linear case. In *NeurIPS*, 2010.
- B. Laurent and P. Massart. Adaptive estimation of a quadratic functional by model selection. *Annals of Statistics*, 28(5):1302–1338, 2000.